

David Pérez Arbués

Engineering Lead · adidas Digital Experiences (DXP)

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// summary

Software engineering leader with 15+ years of experience, currently working within adidas DXP — the engineering organisation that builds and operates adidas.com, the Flagship app, CONFIRMED, and Running. Leading the .com Membership chapter (9 engineers, 5 direct reports), with scope recently expanded to include .com Acquisition (3 engineers focused on Glass decommission) and cross-platform coordination across Upper Funnel surfaces — PDP, PLP, and checkout — through the Latte rendering initiative.

In 2025, stepped up to cover the Director of Software Engineering role during a period of leave — gaining direct exposure to Director-level decision-making and contributing to the 2026 value-led team structure.

Strong hands-on engineering instincts alongside the leadership breadth: in 2026 led AI-enabled engineering transformation — pioneering GenAI-assisted delivery that halved Glass Decommission timelines and earning the adidas Bolder Award (April 2026). A consistent track record of hands-on delivery: critical microfrontend migrations under hard deadlines, internal tooling built before the ecosystem caught up. Grounded in AI/ML research since 2009 (Deep Belief Networks applied to brain-computer interfaces, University of Zaragoza) — a foundation that informs a practical, first-principles approach to AI adoption.

// experience

Senior Software Engineer → Engineering Lead, .com Membership · adidas DXP

Aug 2022 – present

adidas · Zaragoza, Spain

- 2025 — Covered the Director of Software Engineering role during a period of leave: participated in Director-level meetings with the Senior Director of SE, reported across all of Membership (.com + apps), and collaborated on the 2026 value-led team structure (cross-functional squads: backend, .com, iOS, Android)
- Led .com Membership chapter (9 engineers, 5 direct reports) with expanded scope covering .com Acquisition (3 engineers, Glass decommission); responsible for KLO/RUN/Support — including security issue resolution — coaching, capacity planning, and collaboration with QA Excellence on testing standards
- 2026 — Led Glass Decommissioning: delivered in 3 months what was estimated at 6 months with 4.5 FTEs, including unplanned scope — recognised with the adidas Bolder Award (April 2026); led go-live security process (security architecture review, penetration testing coordination); leading AI-driven creation and migration of end-to-end tests across Glass components
- Drove Latte cross-platform initiative: built the microfrontend from scratch and expanded JSON-driven rendering to Upper Funnel surfaces — checkout, PDP, PLP — coordinating joint API contracts and releases across .com and iOS/Android teams
- Internalized a fully external team (12 Merkle contractors → in-house) in 7 months; delivered the resulting microfrontend migration in 3 months on a hard deadline, extending the framework to support nested microfrontends
- Presented microfrontend migration strategy and Latte case study to ~100 adidas engineers each
- Defined and owned testing strategy: coverage thresholds, rotating quality champion role, Sonar gates, githooks, and improved deployment flows — lifting quality culture across both chapters

React TypeScript Node.js XState v5 AWS Lambda DynamoDB Kubernetes Next.js Latte

Tech Lead & Sole Engineer

2024 (parallel to adidas)

Partum · Remote

- Designed and built a new generation smart locker kiosk from scratch — edge-first, Raspberry Pi + NestJS + React inside Docker managed by Balena
- Integrated Kerong smart lock hardware via a custom TCP binary protocol (STX/ETX framed) over raw sockets
- Kiosk UI driven entirely by XState v5 state machines (delivery and pickup flows)
- Moved cloud communication to AWS IoT and WebSockets; provisioned all infrastructure with AWS CDK v2

NestJS TypeScript React XState v5 Balena AWS CDK v2 Raspberry Pi

Software Engineer (Backend Lead, Connection Studio)

Jan 2019 – Aug 2022

adidas · Zaragoza, Spain

- Led the full re-architecture of Connection Studio 2.0: migrated from Java/Spring to Node.js/NestJS/TypeScript, replaced Oracle DB with DynamoDB — serving global adidas.com login and campaign traffic at scale
- Built a suite of internal NestJS libraries before equivalent open-source solutions existed: fluent DynamoDB query builder, decorator-driven Kafka router, @Cache() method decorator, runtime decorator discovery
- Reduced monthly deployments to weekly via GitHub Flow and CI/CD improvements; introduced Sonar, Instana, Grafana, Opsgenie
- Migrated mission-critical microservices to AWS Lambda; introduced Terraform and AWS CDK for IaC
- Security Champion: coordinated security architecture reviews and penetration testing for go-live; led security practices across the team throughout the re-architecture
- Mentored one engineer across 3+ years and 3 technology shifts (Java BE → Node.js BE → React FE)
- Promoted to Senior Software Engineer in 2022

TypeScript · Node.js · NestJS · AWS Lambda · DynamoDB · Apache Kafka · Terraform

Software Engineer

Jan 2014 – Jan 2019

GFT Group · Zaragoza, Spain (client: Deutsche Bank UK)

- Technical area lead for architecture, code quality, and security on the EUS project at Deutsche Bank UK
- 80% reduction in on-call incidents through Java and Unix quality engineering
- Led international knowledge transfer to HCL India (technical and functional, delivered in English); led disaster recovery planning across geographies
- Voluntarily took on frontend work on dbPalace after self-studying Angular when no frontend engineer was available

Java · Spring Boot · Angular · PL/SQL

Consultant / Analyst

May 2011 – Jan 2014

everis · Madrid, Spain (client: Ministerio de Economía y Competitividad)

- Developer and analyst for the unified application processing system and its data model
- Continuous development and improvement of the document processing system
- Created and improved Oracle stored procedures for inter-application data transfers

Java · Oracle SQL · PL/SQL

Researcher

Jan 2009 – Jan 2010

Universidad de Zaragoza · Zaragoza, Spain

- Research on Deep Belief Networks applied to brain-computer interfaces: using EEG signals to control a wheelchair through pattern recognition
- Implemented learning algorithms in Matlab; built a handwritten digit recogniser via camera as proof of concept
- Publication: zaguan.unizar.es/record/5100

Matlab · Deep Belief Networks · EEG signal processing

// skills

TypeScript · JavaScript · Node.js · React · Go · Python · Java · NestJS · XState · Express.js · AWS (Lambda, DynamoDB, API Gateway, S3, CDK, IoT) · Kubernetes · Docker · Terraform · Balena · Apache Kafka · WebSockets · PostgreSQL · Instana · Grafana · Sonar · Scrum (PSM I) · SAFe (Agilist + Practitioner) · Kanban · GitHub Flow · Microfrontend architecture

// education

Universidad de Zaragoza

2000 – 2010

Ingeniería Superior de Telecomunicaciones · Telecommunications Engineering — Master's equivalent (EQF Level 7)

// certifications

Oracle Cloud Infrastructure 2023 Certified Architect Associate

Oracle · 2023

AWS Certified Solutions Architect – Associate

AWS · 2022

CKAD: Certified Kubernetes Application Developer

The Linux Foundation · 2022

AWS Certified Developer – Associate (959/1000)

AWS · 2020

CCA Spark and Hadoop Developer

Cloudera · 2019

Certified SAFe® 4 Agilist

Scaled Agile · 2019

Professional Scrum Master (PSM I)

Scrum.org · 2018

Certified SAFe® 4 Practitioner

Scaled Agile · 2018

Oracle Certified Professional, Java SE 8

Oracle · 2017

Cambridge English: Advanced (CAE) — Grade A — C2

University of Cambridge · 2016

// awards

Bolder Award — AI Native Guild

adidas · April 2026

Selected by the M1 panel for the inaugural Bolder Awards for leading the Glass Decommissioning initiative. Estimated at 6 months with 4.5 FTEs — delivered in 3 months including unplanned scope, by pioneering GenAI-assisted component re-implementation. Built an AI-powered SDLC orchestration platform (Claude Agent SDK + XState v5 + Next.js) that drives tickets autonomously through analysis → implementation → code review → security audit → PR. Learnings shared across DXP and CE teams to drive broader AI adoption.

// languages

Spanish Native **English** C2 (Cambridge Grade A) **German** Basic

// projects

Latte template AI analysis pipeline In progress

Using AI to build agents, commands, and tools that reduce tech debt across Latte templates — surfacing and fixing redundant state patterns, dead content, and hardcoded strings, while driving harmonisation across templates. Goal: lower the cost of creating new templates for markets, reduce the maintenance burden on existing ones, and reduce the risk of production issues caused by misconfiguration.

Claude Code **TypeScript** **Next.js** **Latte**

AI homelab sandbox Active

Proxmox NAS running Ollama, n8n, and Claude in a sandboxed environment for testing AI integrations before bringing them to production. Used to explore advisor patterns, token reduction strategies, prompt injection hardening, and agentic workflow design.

Proxmox **TrueNAS** **Ollama** **n8n** **Docker**

Home automation & IoT fleet Active

Declarative management of OpenBeken/Tuya smart devices via Ansible and Home Assistant. Being replaced by obk-fleet — a Go daemon using the Kubernetes operator pattern (desired state, reconciliation loop, drift detection) for full fleet management.

Go **Ansible** **Home Assistant** **MQTT** **OpenBeken**

Technical writing In progress

Article series on AI for engineers — 'The Good, the Bad and the Ugly' — covering practical AI adoption, honest trade-offs, and engineering perspective on LLMs. Published on Medium and LinkedIn.

AI **LLMs** **GenAI**